

PATRICIA (PATTI) DEGNER

Senior Software Engineer | AWS Infrastructure and Cloud Platforms | Agentic AI

620.899.7677 | pattidegner@gmail.com | linkedin.com/in/patricia-degner | pdegner.github.io | github.com/pdegner

SUMMARY

Senior Software Engineer specializing in AWS infrastructure, Python backend systems, and production-grade agentic AI. Built cloud platforms and data pipelines processing millions of events, established CDK and CI/CD foundations from scratch, and developed evaluated multi-agent systems that safely automate software engineering workflows.

TECHNICAL SKILLS

AWS: CDK, Lambda, Step Functions, S3, Glue, Athena, SageMaker, RDS, Bedrock, CloudWatch, EventBridge, SES, IAM, API Gateway, VPC, CodePipeline, Terraform

Languages: Python, SQL, TypeScript, JavaScript

Data and ML: PySpark, Airflow, PyTorch, Spark SQL, PostgreSQL

Agentic AI: LangGraph, Claude Agent SDK, Model Context Protocol (FastMCP), evaluation harnesses, tool scoping, guardrails, pgvector

Platform: Infrastructure as code, CI/CD with GitHub Actions, automated testing, React and React Admin, cloud cost optimization

EXPERIENCE

Senior Developer, InfoBate - Pittsburgh, PA (remote)

Jun 2023 - Present

- Established the team's AWS CDK codebase in Python from scratch, defining the infrastructure as code patterns and CI/CD deployment pipeline the team uses daily; mentored engineers in CDK best practices.
- Designed, built, and automated production AWS data pipelines supporting daily reporting, bi-monthly analytics, and data ingestion workflows using Python, CDK, Lambda, Glue, SageMaker, S3, Step Functions, and Athena.
- Designed and built an internal seven-node LangGraph state machine using the Claude Agent SDK to convert natural-language requests into GitHub pull requests across three production repos, with human approval gates and tool scoping enforced at the state machine level.
- Reduced LLM token costs of the agentic AI workflow by approximately 25% by routing individual steps to the lowest-cost suitable model, using per-node CloudWatch instrumentation to measure cost and behavior against a 42-test automated evaluation suite.
- Partnered directly with customers to translate business requirements into shipped software. Owned backend development for client-facing features, including PostgreSQL schema design, query optimization, React Admin interfaces, and production database upgrades.
- Redesigned core data tables during onboarding, cutting a high traffic query from 10M+ rows to ~10K.
- Decommissioned an unused RDS database with misconfigured minimum compute after validating no downstream dependencies, eliminating ~\$1,400/month in cloud spend.

Software Development Engineer L4, Amazon - Louisville, CO (hybrid)

May 2022 - Mar 2023

- Scaled and optimized production AWS data pipelines processing over 2 million events daily for Amazon Go, improving reliability and operational efficiency.
- Designed and implemented a resumable Step Functions workflow using Lambda, SES, and API Gateway that eliminated redundant preprocessing and saved ~13 hours of compute per execution.
- Expanded test coverage to 80% of existing and 100% of new code; streamlined internal tooling in TypeScript and React.

Decision Scientist, Ibotta - Denver, CO (remote)

Jan 2021 - Apr 2022

- Automated recurring data collection and analysis in Airflow, saving ~\$36,000 annually.
- Migrated legacy workflows to PySpark and optimized the underlying algorithms, cutting runtime ~5 hours/week.
- Authored how-to guides for Airflow, PySpark, and SQL that cut ramp time for new decision scientists by half.

PROJECTS - Details available at pdegner.github.io

Notes MCP Server (Python, FastMCP)

Built an open-source Model Context Protocol server in Python providing a persistent tagged note store, implementing tools, URI addressable resources, and reusable prompts so context, thoughts, and decisions survive across agent sessions.

- Hardened the server against untrusted model input: regex-validated note IDs to block path traversal, atomic writes to prevent truncation on interrupt, and restrictive file permissions; security covered by 59 evaluation tests and CI.

Climbing Regulations Retrieval System (RAG)

Built and deployed a RAG application answering questions about federal climbing regulations with verifiable, accurate citations.

- Engineered a placeholder-based citation architecture that makes fabricated citations structurally impossible rather than relying on prompting; 100% citation validity across 86 evaluated answers.
- Implemented hybrid retrieval combining pgvector semantic search with Postgres full text search, reranked with Amazon Bedrock Rerank and generated with Amazon Bedrock Converse under a strict citation only prompt.
- Ran a 43-question evaluation across 4 retrieval strategies to quantify tradeoffs, then shipped the system as a serverless AWS stack (serverless API, Lambda, RDS, CDK) for under \$1 total spend.

EDUCATION

University of California, Berkeley - Master of Information and Data Science

Dec 2020

Thesis: Audio Super Resolution using Deep Learning.

- Trained a spectrogram autoencoder in PyTorch on 85GB of audio to reconstruct high frequency content from compressed audio, served through a real-time Flask interface.

University of Kansas - B.S. Supply Chain Management

May 2017